Study document: steps 1 to 4

Data cleaning · Breaking the problem down · Features and scaler · The Random Forest

Stability and Structural Validity of Learned Node Importance in Citation Networks - Phase B. Status as of 31 October 2026. Written so the team can learn what was done, why, and what is still open.

1. The big picture in one page

The question. Papers can be ranked by importance with centrality measures (how connected, how central, how "influential" a paper is). But citation data is incomplete and noisy, and small changes in the network can reorder the ranking. Phase A proposes to combine five measures with a machine-learning model, a **Random Forest**, and asks whether the combined ranking is more stable than any single measure when the network is damaged.

The data. PubMed-Diabetes: 19,717 papers about diabetes and 44,338 directed citations (A cites B = an edge A → B).

The pipeline (each box is one block in the code):

A Features	B Scaler	C Random Forest	D Ranking	E Damage	F Stability	G Core
5 centralities per paper	Min-Max, fitted once, then frozen	learns PageRank from the features	scores → ranks	remove / add citations	how much do ranks move?	important and stable papers

This document covers the first four tasks of the plan: **1** cleaning the data, **2** breaking the problem down, **3–4** writing the features and the Random Forest and training it (your lecturer's task 3 is our steps 3 and 4).

Term	Meaning
Centrality	a number that says how important a paper is because of its place in the network
PageRank	a paper is important if important papers cite it (the idea behind early Google search)
DAG	directed graph without cycles; a citation network should be one, because papers only cite older papers
Min-Max scaling	x' = (x − min) / (max − min): squeezes a feature into [0, 1]
Random Forest	many decision trees, each trained on a random sample of the data; the forest averages their answers
Hold-out test	train on 80 % of the papers, test on the other 20 % the model has never seen
Spearman ρ / Kendall τ	how similar two rankings are (1 = identical order, 0 = unrelated)

2. Step 1 - Cleaning the database

Raw files. The node file holds, for each of the 19,717 papers, a class label (diabetes type 1, 2 or 3: 4,103 / 7,875 / 7,739 papers) and 500 word-frequency columns. The edge file holds 44,338 citations. The raw files are never modified; cleaning writes new files to data/processed.

What "clean" means for a citation graph. Every check below protects a specific later calculation:

Check	Why it matters	Found	Action
Duplicate edges	the same citation counted twice inflates degree and PageRank	0	-
Self-loops (A cites A)	carry no information, distort PageRank	3	removed
Mutual citations (A↔B)	impossible in time: the older paper cannot cite the newer one; creates cycles	11 pairs	one edge of each pair removed
Edges to unknown papers	would create phantom nodes	0	-
Isolated papers	cannot be ranked by structure	0	-
Cycles left afterwards	a citation network should be a DAG	0	verified

Result: 19,717 papers and 44,324 citations; one connected component; no cycles. The 500 word columns were dropped because our model uses only the structure of the graph.

Decisions in this step

- **Which edge of a mutual pair to keep.** The data has no publication years, so the PubMed ID is used as a stand-in for time (IDs grow over time). We keep the edge from the higher ID to the lower ID. **This is an assumption**: afterwards we checked it, and only 89.6 % of all real citations go from a higher to a lower ID. It affects 11 of 44,324 edges, so the risk is small, but it is not a fact.
- **"Validate temporal consistency" (Phase A).** Impossible by year, so it is done structurally: no cycles remain.
- **Dataset.** Phase A names Cora as "preliminary"; we use PubMed-Diabetes, the dataset provided.

Worth knowing. 15,840 papers (80 %) cite nothing inside the dataset. The dataset keeps only references that point to other papers in it, so this is a property of the sample, not an error. It matters later: it strongly affects closeness and PageRank (sections 3 and 5).

3. Step 2 - Breaking the problem into small parts

The Phase A model was split into the seven blocks shown in section 1. Each block is one code module with a clear input and output, so it can be tested alone and replaced without touching the others. For example: the scaler (B) only learns min and max from a table and applies them; it knows nothing about graphs or forests.

Fixing the ambiguous rules before writing code

The real work of this step was finding the places where Phase A does not say precisely what to calculate, and deciding each one explicitly. A wrong guess would quietly change every result. The formula changes are explained in section 6.

#	Ambiguity in Phase A	Our rule
1	Closeness is defined by "distance from a node to all others"; most papers cannot reach most others, so the distance is infinite	closeness measured on incoming paths, with the Wasserman-Faust correction for unreachable papers; a paper nobody reaches gets 0 (section 6, F1)
2	PageRank formula ignores papers with no outgoing citations (80 % of papers)	standard fix: their score is redistributed evenly (F2)
3	k-core is defined for undirected graphs; ours is directed	computed on the undirected version, self-loops removed (F5)
4	After damage, scaled values can leave [0, 1]	the scaler is fitted once on the original graph, never refitted, and values are not clipped (Phase A requirement 12)
5	A paper whose mean score is 0 has no coefficient of variation	its instability is set to 0 (as Phase A says) and these papers are counted separately
6	Many papers have exactly equal scores (ties)	tied papers share the average rank; scores are rounded to 12 significant digits first so rounding noise cannot split a tie
7	Spearman and Kendall formulas assume no ties	tie-aware versions are used (F9, F10)
8	Which scores are used for the coefficient of variation: scaled or raw?	raw ("natural units") (F8)

4. Step 3 - The features and the scaler

For every paper we compute five centralities. **Local:** degree (how many citations it gives and receives), k-core (how deep inside a densely connected group it sits). **Global:** betweenness (how often it lies on the shortest path between two other papers), closeness (how quickly the rest of the network reaches it), PageRank.

All five take well under a second together on the full graph (betweenness and closeness use the igraph library; the others use networkx). Each result is tested against a reference implementation on small graphs.

What the real data showed (none of this was visible from the Phase A text):

- Only **1,807 papers (9 %)** have betweenness above zero. A paper can only lie "between" two others if it both cites and is cited; most papers do only one of the two.
- **2,047 papers** have closeness 0: the never-cited ones (nobody reaches them).
- **Degree and k-core are almost the same ranking** (Spearman 0.97), so the forest gets little new information from having both.

- Closeness and PageRank agree well (Spearman 0.73); degree and PageRank much less (0.29).

The scaler. Min-Max is fitted once on the original graph: $x' = (x - \min_o) / (\max_o - \min_o)$, with $\min_o$ and $\max_o$ taken from the original graph only. On the original graph all values lie in [0, 1]. On a damaged graph the same $\min_o$ and $\max_o$ are used, so a value can fall outside [0, 1]; that is intended, and important for section 5.

5. Step 4 - The Random Forest

What it is and how we set it up

A **decision tree** repeatedly splits the papers by a rule such as "closeness > 0.2?" until the groups are small, and predicts the average target of each final group. One tree is unstable; a **Random Forest** trains many trees on random samples of the papers (bootstrap) and averages their answers. Because every tree can use any combination of features, the forest can capture non-linear interactions between the centralities, which a plain average could not.

Setting	Value	Note
Input	the 5 scaled features (or 4 without PageRank)	two versions, see below
Target	PageRank of the paper on the original graph	Phase A; scaled with the same Min-Max
Trees	200	no tuning on purpose: the question is about stability, not best fit
Minimum papers per leaf	5	keeps the saved model at 18–33 MB and avoids memorising single papers
Training set	all 19,717 papers	as in Phase A; one forest, trained once, never retrained on damaged graphs
Check	80 / 20 hold-out	a sanity check only; the experiments use the forest trained on all papers
Randomness	seed 42	same seed → identical forest (verified on a fresh clone of the repository)

Results

	Hold-out R^2	Hold-out Spearman	What the forest relies on
Forest **with** PageRank as an input	0.999	1.000	PageRank: 99.8 % of the feature importance
Forest **without** PageRank	0.75	0.80	closeness 0.47, degree 0.44, k-core 0.05, betweenness 0.05

For comparison, the best single centrality (closeness) follows PageRank with Spearman 0.73; k-core 0.37, degree 0.29, betweenness 0.19. So the forest without PageRank combines the features into something closer to PageRank than any one of them, which is exactly what the model is meant to do. In-sample Spearman is 0.81 against 0.80 on held-out papers: there is almost no overfitting.

The surprises

1. With PageRank as an input, the forest only copies PageRank

Phase A uses PageRank both as a feature and as the target. The forest discovers the shortcut at once: 99.8 % of its decisions use the PageRank column, and $R^2 = 0.999$ looks excellent but means nothing. This model learned nothing about the network. **Consequence:** we train a second forest without PageRank as an input and treat it as the main model; the first stays in the study as the Phase A design (decision D2).

2. Only closeness and degree really matter, and the forest has no "feature randomness" (new, not yet decided)

Without PageRank, 90 % of the importance is carried by closeness and degree. "Combining five measures" is in practice combining two.

Phase A's hypothesis H2 says the forest is robust "because it utilizes feature randomness" (each split looks at a random subset of the features). **Our forest does not do this.** In scikit-learn the default for regression forests is max_features = 1.0: every tree considers all features at every split, so only the random sampling of papers differs between trees. We did not change the default, and we did not notice until writing this document. Hold-out accuracy hardly changes if the setting is changed (without PageRank: R^2 0.747 with all features, 0.746 with half of them), so this is a design question, not an accuracy problem. **Open decision, see section 8.**

3. Trees cannot extrapolate (matters from step 7 on)

A tree answers a new value with the answer of the nearest group it saw in training. After damaging the graph, many scaled feature values leave [0, 1] (the scaler is frozen), and the forest gives all of them the same edge answer. Such papers will look "stable" only because the model can no longer tell them apart. We measure how often this happens and report it as a limitation.

6. Formula changes compared with Phase A

Phase A's formulas are quoted as written (Appendices B and C and Chapter 4). N is the number of papers, d(u,v) the length of the shortest citation path from u to v. **Six formulas differ in substance (F1, F2, F8, F9, F10, F12); the others only differ in how they are written or are identical.** Items that need your advisor's approval are marked ▶ and listed again in section 8.

	Formula	Status
F1	Closeness	▶ CHANGED (direction and unreachable papers)
F2	PageRank	▶ extended (papers with no outgoing citations)
F3	Degree	identical
F4	Betweenness	rescaled; no effect on any result
F5	k-core	equivalent; directed → undirected
F6	Min-Max scaling	identical (frozen, not clipped)
F7	Target and forest score	linear rescaling; no effect on rankings
F8	Instability (CV)	▶ CHANGED (units; population std)
F9	Spearman ρ	▶ CHANGED (tie-aware)
F10	Kendall τ	▶ CHANGED (τ-a → τ-b)
F11	Ranking stability and instability	identical
F12	Network core	▶ CHANGED (second, adjusted rule added)
F13	Perturbation	identical, with constraints; one variant added

F1 - Closeness ▶

Phase A: the average shortest-path distance from a node to all other nodes:

$$\mathrm{CL}(v) = (N - 1) \,/\, \Sigma_{y \neq v}\, d(v, y)$$

Problem: (a) in a citation DAG a paper can only reach the older papers it cites, directly or indirectly. 15,840 papers (80 %) cite nothing in the dataset, so their sum is empty; (b) the sum is infinite whenever some paper cannot be reached, which is true for almost every paper.

Implemented: the **reverse direction** (how quickly the rest of the network reaches v) with the Wasserman-Faust correction. Let R(v) be the set of papers that can reach v through citation chains, r = |R(v)|:

$$\mathrm{CL}(v) = [\, r \,/\, (N - 1) \,] \cdot [\, r \,/\, \Sigma_{u \in R(v)}\, d(u, v) \,] \qquad \text{and } \mathrm{CL}(v) = 0 \text{ if } r = 0$$

Reading it: the second bracket is the inverse of the average distance to the papers that can reach v; the first bracket is the share of the network that can reach v and punishes papers few can reach. When every paper can reach v (r = N − 1) the formula becomes Phase A's (N − 1)/Σd, only with distances measured towards v.

Effect (measured): papers with closeness 0: **2,047** (the never-cited) with our direction against **15,840** with Phase A's. Spearman with PageRank: **+0.73** against **−0.27**; the two versions correlate at −0.29, so they are nearly opposite measures. The forest without PageRank is equally accurate with either (hold-out R^2 0.747 vs 0.691, Spearman 0.797 vs 0.801).

Why: with Phase A's direction closeness mostly measures how deep a paper's own reference list reaches, not how important it is. Phase A's test plan also asks for 0 when no finite path exists, which our version satisfies.

F2 - PageRank ▶

Phase A:

$$\mathrm{PR}(v) = (1-\alpha)/N + \alpha \cdot \Sigma_{u \in \mathrm{in}(v)}\, \mathrm{PR}(u) \,/\, \deg_{\mathrm{out}}(u), \quad \alpha = 0.85$$

Problem: a paper with no outgoing citation (a "dangling" paper) passes its score to nobody, so score disappears at every iteration and the values no longer sum to 1. With 80 % dangling papers this is not a rounding detail.

Implemented (the standard networkx rule): dangling papers D spread their score evenly over all papers:

$$\mathrm{PR}(v) = (1-\alpha)/N + \alpha \cdot [\, \Sigma_{u \in \mathrm{in}(v)}\, \mathrm{PR}(u) \,/\, \deg_{\mathrm{out}}(u) \;+\; (1/N)\, \Sigma^{d \in D}\, \mathrm{PR}(d) \,]$$

Details: edges point from the citing to the cited paper, so cited papers collect score; α = 0.85; iteration stops when the total change is below $N \cdot 10^{-6}$ (networkx default); the scores sum to 1 (tested). The formula changes exactly where Phase A is silent, so this is an extension, not a contradiction.

F3-F7 - Unchanged or equivalent

F3 Degree: $C_D(v) = (\deg_{in}(v) + \deg_{out}(v)) / (N-1)$, exactly as in Phase A (it can exceed 1, because it counts both directions).

F4 Betweenness: Phase A: $BC(v) = \Sigma\, \sigma_{st}(v)/\sigma_{st}$ (unnormalised). We divide by (N−1)(N−2), the largest possible value in a directed graph, and compute it exactly (the sampled version we first considered was too inaccurate: Spearman 0.55 with the exact one). Dividing by a constant changes neither the Min-Max-scaled value nor the coefficient of variation, so **no result is affected**.

F5 k-core: coreness $k(v) = \max\{k : v \in H_k\}$ computed on the undirected graph without self-loops. Phase A then normalises with $(k - k_{min})/(k_{max} - k_{min})$; our global Min-Max does the same ($k_{min} = 1$, $k_{max} = 10$ on the original graph). One subtlety: on a damaged graph we keep k_{min} and k_{max} of the original (frozen scaler), while Phase A's wording "values present in the network" could be read as recomputing them.

F6 Min-Max: $c_i'(v) = (c_i(v) - \min c_i) / (\max c_i - \min c_i)$, min and max from the original graph only; no clipping.

F7 Target: Phase A: y(v) = PR(v). We train on the Min-Max-scaled PageRank, $y' = (PR - PR_{min})/(PR_{max} - PR_{min})$, and convert the forest's output back ($PR = y' \cdot (PR_{max} - PR_{min}) + PR_{min}$) before computing instability. A linear change of the target leaves the trees' splits and every ranking unchanged.

F8 - Instability (coefficient of variation) ▶

Phase A:

$$\mathrm{InStability}(v) = \sigma(SC_v) \,/\, \mu(SC_v) \quad \text{if } \mu(SC_v) \neq 0, \text{ otherwise } 0$$

Implemented: the same ratio with (a) σ as the **population** standard deviation (divide by K, not K − 1; Phase A does not say) and (b) the condition μ > 0 (scores are never negative); (c) **computed on natural units**: the raw centrality value for single measures, and the forest's score converted back to PageRank units (F7).

The real change is (c). Phase A says features are Min-Max-normalised "to preserve a meaningful scale for CV computation", i.e. it intended the CV on normalised values. We do not, because the CV is not invariant to a shift: CV(x + c) ≠ CV(x). Min-Max subtracts the minimum, so every paper at the minimum gets 0 (mean 0, CV forced to 0 although it may fluctuate), and after damage the frozen scaler can produce negative values, making a mean near zero and the CV explode or change sign. In natural units the CV is always defined and meaningful. For the forest nothing changes if Phase A's target is the raw PageRank (it is already in PageRank units); for the single-centrality baselines it matters. **Needs approval.**

F9 and F10 - Spearman and Kendall with ties ▶

Phase A:

$$\rho = 1 - 6\,\Sigma\, d_i^{\,2} \,/\, (\, n\,(n^2 - 1)\,) \qquad \tau = (n_e - n^d) \,/\, (\, n\,(n-1)\,/\,2\,)$$

Problem: both formulas are exact only when no two papers share a rank. Here they do, massively (e.g. 83 % of paper pairs are tied for betweenness).

Implemented: **Spearman** = the Pearson correlation of the average ranks; **Kendall τ-b**, which divides by the number of pairs that are not tied in either ranking:

$$\tau\text{-b} = (n_e - n^d) \,/\, \sqrt{(\,(n_0 - n_1)(n_0 - n_2)\,)}, \quad n_0 = n(n-1)/2, \quad n_1, n_2 = \text{tied pairs in each ranking}$$

Effect (measured, one run, 10 % of citations removed):

Measure	Spearman: ours	Spearman: 6Σd² formula	Kendall τ-b (ours)	Kendall τ-a (Phase A)	pairs tied
k-core	0.947	0.953	0.914	0.657	29 %
degree	0.957	0.961	0.928	0.698	25 %
betweenness	0.974	0.994	0.968	**0.166**	83 %
PageRank	0.929	0.929	0.858	0.839	1.5 %

Reading it: τ-a would report that betweenness is nearly unrelated to itself after 10 % removal, only because most of its pairs are tied. That is a property of the formula, not of the ranking. Phase A itself remarks that Kendall is informative "when rankings contain ties", which is the situation τ-b is designed for. **Needs approval.**

F11-F13

F11 Ranking stability: $\mu(CR) = (1/K)\,\Sigma\, cr_k$ and $I_{rank} = \sigma(CR)/\mu(CR)$, unchanged (population σ).

F12 Network core ▶: Phase A: Core = { v : InStability(v) ≤ ε }, with ε the 10th percentile of all papers' instability, together with high importance. We keep this rule (importance = top 10 % of the base ranking) **and** add an adjusted one, because the first fails on this data:

$$\text{Phase A rule:} \quad \text{core} = \{\, \text{rank}(v) \le 0.1N \;\text{ and }\; CV(v) \le P_{10} \text{ of CV over all papers} \,\}$$
$$\text{Adjusted rule:} \quad \text{core} = \{\, \text{rank}(v) \le 0.1N \;\text{ and }\; CV(v) \le \text{median of CV over the top-10 \% papers} \,\}$$

Why: under edge removal more than 10 % of papers never change score, so $P_{10} = 0$ and only papers at the very bottom of the ranking (average base rank about 18,700 of 19,717) qualify: the core is empty for the forest, PageRank and closeness. The adjusted rule judges stability among the important papers only. **Needs approval.**

F13 Perturbation: $E_p = E \setminus E_r$ (remove p % of the edges) and $E_p = E \cup E_n$ (add p %), as in Phase A, with n = round(p · |E|) edges (p = 5, 10, 20 % gives 2,216 / 4,432 / 8,865), sampled without repetition; added edges are never self-loops or copies of existing edges; the node set never changes. We add a second addition variant in which every new edge goes from the earlier to the later paper in a topological order of the cleaned graph, so no citation cycle can appear (random edges make groups of up to 2,498 papers that all reach each other).

7. Problems found, and how they were solved

Problem	How it was found	Solution
Sampled betweenness (first plan) was inaccurate	timed and compared with the exact value: Spearman 0.22 / 0.55 / 0.63 for 100 / 500 / 1,000 sampled sources; finds only 535 of the 1,807 non-zero papers	exact betweenness with igraph: under 1 s instead of 161 s
PageRank as input makes the forest a copy of PageRank	feature importance 0.998 on the PageRank column	second forest without it
Phase A formulas undefined or misleading on this data	applying them to the real graph (F1, F2, F9, F10)	section 6
Rounding noise (~10^{-16}) split true ties at random	the later pipeline did not reproduce the step 5 ranking exactly	scores rounded to 12 significant digits before ranking (smallest real gap between PageRank values: ~10^{-8})
Forest prediction on several threads was not bit-reproducible	same seed, scores differed in the 16th digit	prediction on one thread; rankings were never affected

8. Decisions: approved by us, to approve with the advisor, and still open

A. Need the advisor's approval (they change Phase A)

#	Change	Our reasoning	Alternative
1	PubMed-Diabetes instead of Cora	the dataset provided	find Cora
2	Two forests; the one **without** PageRank is the main model	with it, the forest copies PageRank (99.8 %)	keep only the Phase A design and say it is trivial
3	Closeness on incoming paths, Wasserman-Faust (F1)	Phase A's version is 0 for 80 % of papers and negatively related to PageRank (−0.27)	outgoing closeness as written
4	Instability on natural units (F8)	CV is not shift-invariant; Min-Max values can be 0 or negative	CV on normalised values as Phase A intended
5	Tie-aware Spearman and Kendall τ-b (F9, F10)	formulas of Phase A break with 25-83 % tied pairs	Phase A formulas, with random tie-breaking
6	Second, adjusted core rule (F12)	Phase A rule gives an empty core under removal	report only the Phase A rule
7	Exact instead of sampled betweenness	sampling was inaccurate and adds noise	sampling

B. Decided by us, to inform the advisor

- Mutual citations resolved with "higher PubMed ID = newer" (11 edges; holds for 89.6 % of citations).
- PageRank dangling papers redistribute evenly (F2); k-core on the undirected graph (F5); population standard deviation (F8).
- Forest: 200 trees, minimum 5 papers per leaf, seed 42, no tuning; trained on all papers as in Phase A, with an 80/20 hold-out as a sanity check.

C. Still open

Feature randomness in the forest (found while writing this document)

Phase A's H2 assumes the forest uses a random subset of features at each split. Ours uses all features at every split (scikit-learn default). The H2 verdict ("not supported") therefore tested a forest **without** the mechanism the hypothesis names. **Options:** (1) keep as is and say so; (2) retrain with a random subset of features (for example max_features = 0.5) and rerun the experiment (about 10 minutes of computing); (3) report both. **Recommendation: option 3**, because it either confirms or changes the H2 verdict with evidence. Hold-out accuracy is unaffected (R^2 0.746 vs 0.747).

- **Closeness direction (item 3 above).** I chose the incoming direction; the team has not confirmed it. If the advisor prefers Phase A's direction, the forest accuracy is the same but the meaning of the feature changes, and everything from step 3 on must be rerun (about 10 minutes of computing in total).
- **Training and testing on the same papers.** As in Phase A the forest is trained on all papers and the stability experiments use those same papers. This is acceptable for a stability study (no generalisation claim is made), the hold-out check shows little overfitting, but a reader may object. A sentence in the report is enough.